



\$\$\$ \textsf{\huge National Autonomous University of Mexico} \\\ \$\$\$

\$\$\$ \textsf{\huge Faculty of Engineering} \$\$\$

Hola!

This notebook is part of a series of geoscience applications designed to demonstrate practical approaches for solving problems using computational methods. It serves as a hands-on guide for understanding key concepts in energy-related subsurface systems through applied problem-solving rather than purely theoretical development. The focus is on building intuition and technical skills for modeling and analysis in real-world contexts, particularly within subsurface energy systems. These include petroleum reservoirs, geothermal resources, carbon storage and sequestration, and mineral exploration systems. By integrating fundamental principles with computational tools, the series aims to support learners in bridging the gap between geoscience theory and engineering practice, enabling more effective analysis and decision-making in complex subsurface environments.

Authors:

Dr. Moisés Velasco-Lozano

\$\$\$ \textsf{\huge Bubble-point pressure } p_b\$\$\$

1. Introduction

The bubble-point pressure, p_b is the pressure at which the first bubble of free gas forms as the pressure of a liquid hydrocarbon decreases while the temperature remains constant. At the bubble point, the oil contains the maximum amount of gas that can remain dissolved at that temperature. Therefore, the system is at the boundary between a single-phase liquid region and a two-phase region consisting of liquid and gas. When the pressure decreases below p_b , part of the previously dissolved gas begins to come out of solution, resulting in two phases in equilibrium:

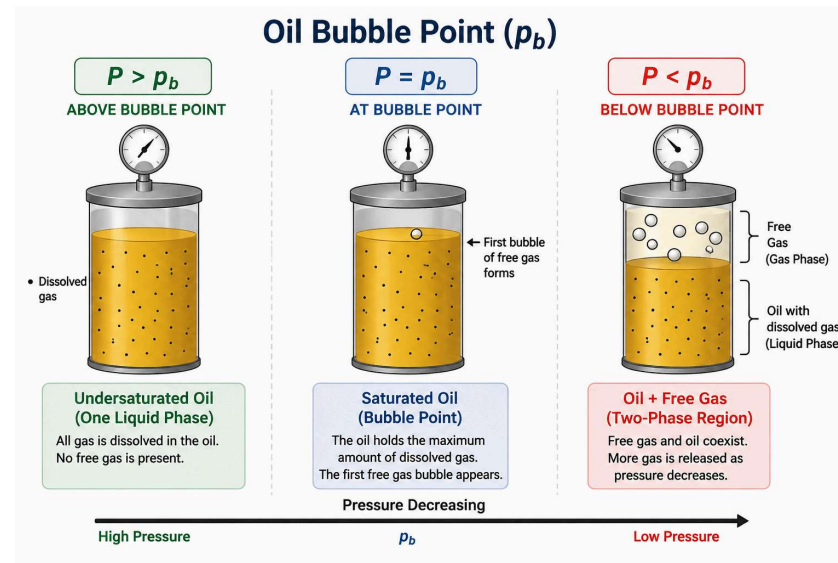
- Liquid phase: oil containing dissolved gas.
- Gas phase: natural gas that has been released from the oil and exists as free gas.

At pressures greater than p_b , the fluid remains as a single liquid phase and is referred to as undersaturated oil. At $p = p_b$, the oil is saturated with gas and is precisely at the condition where gas begins to evolve from solution. At pressures below p_b , oil and free gas coexist, and the amount of liberated gas increases as the pressure decreases.

The bubble-point pressure depends on several fluid properties and conditions, including hydrocarbon composition, temperature, solution gas–oil ratio, and API gravity:

$$p_b = f(y_i, T, R_s, \gamma_{API}, \dots)$$

The following figure shows the behaviour of oil samples at different pressure values:



The bubble-point pressure, p_b , can vary considerably among crude oils because it depends primarily on fluid composition, temperature, and the amount and nature of the gas dissolved in the oil. Approximate ranges often encountered in petroleum systems are:

- Heavy oil | $< 22^\circ$ API | $\sim 100\text{--}1,000$ psi | Usually low p_b limited dissolved gas
- Medium oil | $22\text{--}31^\circ$ API | $\sim 500\text{--}2,000$ psi | Intermediate p_b
- Light oil | $> 31^\circ$ API | $\sim 1,000\text{--}4,000+$ psi | Frequently higher p_b , particularly for volatile oils

There are several numerical correlations available to compute the oil formation volume factor, p_b , some of the most widely used include:

- Standing
- Lasater
- Vázquez

2. Numerical solution

$$p_b = 18.2(F-1.4) \left(\frac{R_{sb}}{\gamma_g} \right)^{0.83} 10^{0.00091T - 0.0125\gamma_o}$$

This correlation is recommended under the following criteria:

- Temperature $100\text{--}258$ $^\circ\text{F}$
- Oil formation volume factor $1.024\text{--}2015$ [bl/bl]
- Solution gas-oil ratio $20\text{--}1425$ [ft³/bl]
- API gravity $16.5\text{--}63.8$

2.1 Import libraries

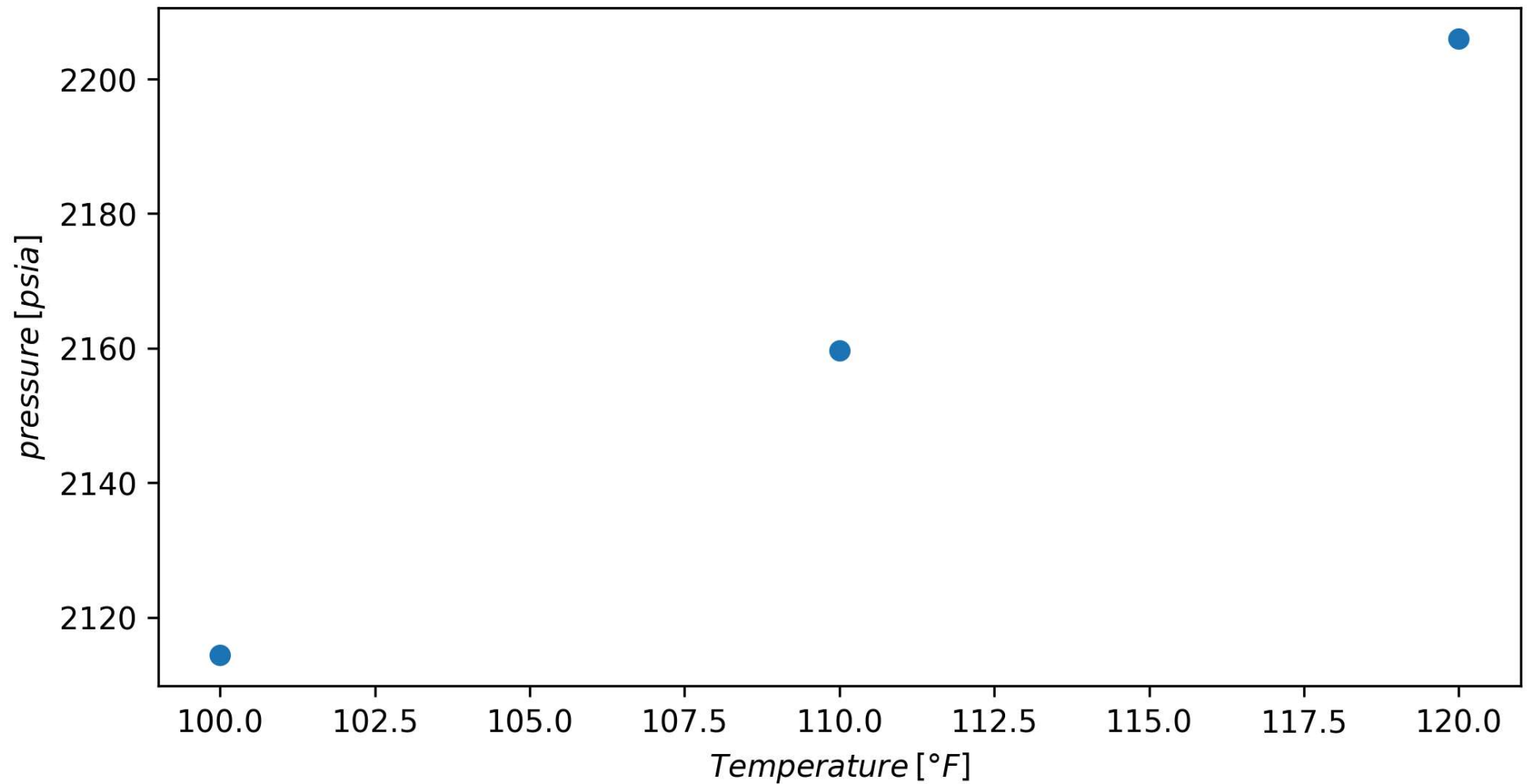
```
In [2]: import numpy as np
import matplotlib.pyplot as plt
```

2.2 Inputs

```
In [30]: gamma_g=0.95  
gamma_o=31  
T=np.array([100,110,120])  
  
Rsb=675
```

2.3 Calculation and plotting

```
In [33]: plt.figure(dpi=300, figsize=(8,4))  
  
# Properties at bubble point  
F=(Rsb/gamma_g)**0.83*10**(0.00091*T-0.0125*gamma_o)  
pb=18.2*(F-1.4)  
  
plt.plot(T, pb, 'o' , linewidth=2)  
  
plt.xlabel('$Temperature\text{:} [^{\circ}F]$')  
plt.ylabel('$pressure\text{:} [psia]$')  
# plt.grid(True)  
# plt.legend()  
plt.show()
```



3. Conclusions and highlights

In []:

4. Comments

I hope this notebook is useful in your journey of coding applied to petroleum engineering and geosciences. I am a researcher interested in Machine learning for subsurface energy systems including oil & gas, carbon capture and storage, geothermal, and minerals. Please reach out!

Dr. Moisés Velasco-Lozano

Associate Professor

Petroleum Engineering Department

National Autonomous University of Mexico (UNAM)

mvelasco@ingenieria.unam.edu

mvelasco@unam.mx

In []: